

FSOC MOBILE TERMINAL COARSE ALIGNMENT PERFORMANCE REPORT

OFFICIAL BENCHMARK VERIFICATION PROTOCOL • PROBLEM STATEMENT 4

Report ID: REP-20261007-93038	Experiment ID: EXP-20261007-93038	Timestamp: 2026-10-07T17:10:38.223670Z	Mode: Simulation Testbench
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1. EXECUTIVE SUMMARY & KPIS

AVERAGE TRACKING ERROR 2.34 px Req: ≤ 10.0 px (RMSE: 2.59px)	LOCK RETENTION RATE 100.0 % Target Loss: 0.0%	ACQUISITION TIME 0.00 s Req: ≤ 2.0 s (Re-acq: 0.00s)	PROCESSING THROUGHPUT 23.8 FPS Latency: 3.33 ms
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2. OFFICIAL REQUIREMENTS VERIFICATION DASHBOARD

OFFICIAL PARAMETER	REQUIRED THRESHOLD	ACTUAL MEASURED	COMPLIANCE MARGIN	STATUS
Acquisition Time	≤ 2.0 s	0.00 s	+2.00 s	PASS
Tracking Error	≤ 10.0 px	2.34 px	+7.66 px	PASS
Target Loss	< 5.0 %	0.00 %	+5.00 %	PASS
Re-acquisition	≤ 1.0 s	0.00 s (Continuous Lock)	+1.00 s	PASS
Processing Speed	≥ 20.0 FPS	23.8 FPS	+3.80 FPS	PASS

3. SUBSYSTEM ARCHITECTURE & HARDWARE CONFIGURATION

Optical Camera: Monochrome Focal Plane Array (640x480, 4.0° x 3.0° FOV)
Target & Motion: Square spot (10px) moving in Figure of 8 at 40.0 px/s
Detection Pipeline: AI + Kalman with clutter rejection enabled
Tracking & Control: 4D Kalman Filter + 2-Axis Gimbal PID (max 5°/s slew limit)
Disturbance Environment: None noise + Clear atmosphere